

FEODOSIA, Ukraine

WMO: 339760

Lat: 45.042N

Lon: 35.382E

Elev: 85

StdP: 14.65

Time Zone: 3.00 (E03)

Period: 82-15

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	11.4	16.1	4.7	7.0	13.1	9.1	8.8	17.0	27.6	22.5	25.3	28.4	10.4	320	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	14.9	87.9	71.3	85.7	70.6	83.5	69.8	74.0	83.5	72.5	82.3	71.2	81.1	6.5	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
70.8	114.0	80.3	69.1	107.3	78.9	67.5	101.4	77.4	37.6	83.3	36.2	82.5	35.1	81.3	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.7	19.3	15.6	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	54.2	38.0	33.2	41.1	51.2	62.3	69.3	74.9	75.0	65.7	55.8	44.4	38.3		
	DBStd	16.20	8.49	10.60	7.87	5.19	6.44	4.62	4.55	4.48	5.87	7.87	9.84	8.42		
	HDD50	1795	374	474	285	42	3	0	0	0	1	35	213	368		
	HDD65	4831	838	891	740	415	129	13	1	1	61	297	618	827		
	CDD50	3344	1	3	11	78	384	579	771	776	473	216	46	6		
	CDD65	904	0	0	0	1	45	142	307	312	83	14	0	0		
	CDH74	5951	0	0	0	2	163	656	2329	2431	346	24	0	0		
	CDH80	1500	0	0	0	0	15	117	635	696	36	1	0	0		
	WSAvg	6.7	7.9	7.6	7.2	6.4	6.1	5.8	6.1	5.8	6.0	6.6	7.1	7.4		
	PrecAvg	17.5	1.4	1.2	1.2	1.4	1.4	1.7	1.4	1.9	1.4	1.0	1.5	1.9		
(15) Precipitation	PrecMax	25.6	4.0	3.0	4.3	3.3	4.8	9.1	4.8	7.2	6.0	3.1	4.1	4.2		
	PrecMin	10.4	0.0	0.2	0.2	0.0	0.0	0.1	0.1	0.0	0.0	0.2	0.1	0.1		
	PrecStd	3.9	0.9	0.7	0.9	0.9	1.1	1.6	1.1	1.6	1.5	0.7	0.9	1.1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.7	56.1	61.9	71.1	81.7	86.9	90.0	91.5	83.6	76.9	66.2	59.0		
		MCWB	48.4	49.5	52.2	57.3	66.0	71.5	73.4	72.0	69.8	66.0	60.6	54.8		
	2%	DB	51.6	51.9	56.6	66.1	77.7	83.1	87.6	88.8	80.6	72.8	62.3	54.3		
		MCWB	47.5	47.4	48.9	55.5	65.1	70.1	71.6	71.1	69.1	64.7	57.6	49.8		
	5%	DB	50.0	49.0	54.0	62.3	74.7	80.4	85.8	86.2	78.2	69.5	59.1	51.4		
		MCWB	46.2	45.6	47.9	53.8	63.1	68.6	70.8	70.5	67.0	63.1	54.6	48.0		
	10%	DB	48.5	46.5	51.3	59.5	72.1	77.6	83.7	83.7	75.6	66.5	56.6	49.1		
		MCWB	45.3	43.6	46.5	52.5	62.3	67.0	70.1	69.7	65.7	60.8	52.9	46.3		
	0.4%	WB	49.1	50.2	53.0	59.3	68.2	73.7	76.9	75.9	72.8	67.9	61.8	55.3		
		MCDB	52.2	55.3	60.2	68.9	78.4	83.3	87.2	87.1	80.7	75.0	65.2	58.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	47.7	47.9	50.1	56.6	66.3	71.8	74.6	73.4	70.4	65.9	58.1	50.5		
		MCDB	50.7	51.6	55.6	63.8	75.1	81.0	83.7	83.8	78.5	71.8	61.6	53.8		
	5%	WB	46.8	45.9	48.5	54.8	64.7	70.0	72.9	72.2	68.2	63.8	55.2	48.3		
		MCDB	49.6	48.8	52.9	60.8	72.7	78.0	82.1	82.9	76.5	68.6	57.9	51.1		
	10%	WB	45.7	43.9	46.9	53.0	63.1	68.4	71.4	70.9	66.4	61.3	53.4	46.5		
		MCDB	48.2	46.5	51.0	58.5	70.8	75.8	81.3	81.7	74.0	66.0	56.2	49.1		
	5% DB	MDBR	8.7	9.7	9.9	10.6	12.7	12.7	14.0	14.9	14.1	12.0	8.7	8.6		
		MCDBR	8.5	10.9	13.5	14.9	15.6	16.1	16.3	17.0	16.4	14.0	11.8	9.5		
		MCWBR	6.3	8.4	8.7	8.6	8.5	8.5	8.1	8.2	9.0	8.5	8.0	7.4		
		MCWBR	8.4	10.5	12.4	13.3	14.3	13.7	14.6	15.2	15.0	12.9	10.4	9.2		
(35) Mean Daily Temperature Range	5% WB	MCWBR	6.6	8.4	8.4	8.3	8.5	7.8	8.4	8.6	9.5	8.4	7.8	7.1		
	taub		0.348	0.377	0.406	0.440	0.412	0.415	0.436	0.438	0.411	0.383	0.359	0.347		
		taud	2.329	2.249	2.195	2.135	2.310	2.343	2.296	2.277	2.320	2.356	2.359	2.331		
(42) Clear-Sky Solar Irradiance	Ebn at Noon		236	250	259	260	272	272	264	259	255	246	230	222		
		Edn at Noon	26	33	40	46	40	39	40	40	35	30	25	23		
	RadAvg		354	612	1020	1559	1964	2111	2147	1891	1374	840	459	314		
(45) All-Sky Solar Radiation	RadStd		41	76	90	154	165	143	107	108	106	81	52	41		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (5 neighbors)	+1.31	N/A	N/A	+1.82	N/A	N/A	-212	-308	+281	+167

Nomenclature: See separate page